EXERCISE 7

Simple Pendulum Energy Conservation and Small Oscillations

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Energy First Integral • Pendulum Isochronism

PHASE 1: STATEMENT & SETUP

Consider a so-called simple pendulum whose shaft has length L and negligible mass, and a suspended mass m at its end. If the pivot is at the origin of coordinates, show that:

1. The gravitational potential energy is $V(\theta) = mgL(1 - \cos\theta)$, where θ is the angle with the downward vertical.
2. The total energy is $E = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos\theta)$.
3. The equation of motion reads $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta$, obtained by requesting $\frac{dE}{dt} = 0$.
4. By considering small oscillations around the vertical, the equation becomes $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$. Show that the general solution reads $\theta(t) = M\cos(\omega t - \phi)$ and relate ω with system parameters. Does ω depend on mass m? Does it depend on amplitude?

PHASE 2: FUNDAMENTAL FORMULATION

Let pivot be at $(0, 0)$ and positive y upwards. Position of bob: $x = L\sin\theta,\ y = -L\cos\theta$. Velocity along arc: $v = L\dot{\theta}$. Total mechanical energy $E = T + V$. In the absence of dissipative friction, mechanical energy is conserved: $\frac{dE}{dt} = 0$.

METHOD OF THE CLASS NOTES [NOTES, P. 5]

The theorem of the notes applies to the full pendulum $\ddot{\theta} = f(\dot{\theta}, \theta) = -\frac{g}{L}\sin\theta$: $f, \partial f/\partial\theta = -\frac{g}{L}\cos\theta$ and $\partial f/\partial\dot{\theta} = 0$ are continuous everywhere, so each pair $\theta(0), \dot{\theta}(0)$ gives one solution.

The small-oscillation equation is $x'' + px' + qx = 0$ with $p = 0, q = g/L$: $k^2 + g/L = 0$, Case 03 with $\rho = -p/2 = 0$ and $\omega = \frac{\sqrt{4q - p^2}}{2} = \sqrt{g/L}$, so $\theta = A\cos\omega t + B\sin\omega t$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Part (i): Potential Energy $V(\theta)$

Vertical height above the lowest point $\theta = 0$ ($y_{\min} = -L$):

$$h(\theta) = y(\theta) - (-L) = -L\cos\theta + L = L(1 - \cos\theta)$$

$$\mathbf{V(\theta) = mgh = mgL(1 - \cos\theta)}$$

Part (ii): Total Energy E

Kinetic energy of point mass: $T = \frac{1}{2}mv^2 = \frac{1}{2}m(L\dot{\theta})^2 = \frac{1}{2}mL^2\dot{\theta}^2$.

$$\mathbf{E = T + V(\theta) = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos\theta)}$$

Part (iii): Equation of Motion via Energy Conservation

Differentiating $E(t)$ with respect to time using the chain rule:

$$\frac{dE}{dt} = \frac{1}{2}mL^2(2\dot{\theta}\ddot{\theta}) + mgL(\sin\theta \cdot \dot{\theta}) = mL^2\dot{\theta}\ddot{\theta} + mgL\dot{\theta}\sin\theta = 0$$

Factoring out $mL^2\dot{\theta} \neq 0$:

$$mL^2\dot{\theta}\left(\ddot{\theta} + \frac{g}{L}\sin\theta\right) = 0 \implies \mathbf{\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta}$$

Part (iv): Small Oscillations Linearization & Frequency Analysis

For $|\theta| \ll 1$, $\sin\theta \approx \theta$. The ODE simplifies to:

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \iff \ddot{\theta} + \omega^2\theta = 0, \quad \text{where } \omega = \mathbf{\sqrt{\frac{g}{L}}}$$

The general solution is $\theta(t) = c_1\cos\omega t + c_2\sin\omega t$. In amplitude-phase form:

$$\theta(\mathbf{t}) = \mathbf{M}\cos(\omega\mathbf{t} - \phi)$$

PHASE 4: CONCEPTUAL ANSWERS & MASTER RESULTS

NATURAL FREQUENCY & ANSWERS TO QUESTIONS

$$\boxed{\omega = \sqrt{\frac{g}{L}} \quad [\text{rad/s}], \qquad T_0 = 2\pi\sqrt{\frac{L}{g}} \quad [\text{s}]}$$

1. Does frequency depend on mass m? NO. Mass m cancels identically from both terms in the equation of motion.

2. Does frequency depend on amplitude M? In the *linear small-amplitude approximation*, **NO** (Galileo's isochronism). In the *exact nonlinear pendulum*, the period lengthens with amplitude according to complete elliptic integrals: $T(M) \approx T_0(1 + \frac{1}{16}M^2 + \dots)$.